

Kushal Singh

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Summary

Staff Software Engineer with 10 years building distributed, cloud-native platforms — twice as a founding engineer carrying systems from zero-to-one through production scale. Shapes architecture for high-throughput distributed systems handling tens of millions of events per hour, raises the production bar through progressive delivery and observability, and multiplies team output through mentorship and platform thinking. Currently working at the intersection of Agentic AI and data-intensive systems.

Experience

Senior Member of Technical Staff — Salesforce

August 2021 – Present | Bengaluru

- **Asset Inventory Platform:** One of the founding engineers on the ingestion and serving layer now processing **600K asset/event records per minute**; shaped the platform's architecture from inception, led it through its largest reliability and scale milestones, and drove cross-team design reviews with upstream producers and downstream consumers to establish the data contracts and SLOs the platform runs on today.
- **Distributed Systems & Performance:** Diagnosed the serialization bottleneck at the core of the ingestion path and led a migration to an event-driven, distributed model with multi-threaded workers and fine-grained transaction boundaries — unblocking concurrent multi-job execution and materially increasing throughput without a hardware footprint change.
- **Agentic AI Integration:** Embedded agentic AI capabilities into the service ecosystem by building custom MCP servers and skills, directly improving developer productivity and reducing manual workflows.
- **Production-safe automation library:** Co-developed an internal Python library to automate day-to-day operational tasks in a secure, auditable manner — enforcing compliance, eliminating manual changes, and adding a programmatic security layer.
- **Observability & Reliability Engineering:** Established end-to-end production confidence by building an integration test framework, instrumenting services with APM, standardizing API documentation via Swagger, and enabling progressive delivery through Argo Rollouts — collectively raising system reliability, resilience, and observability.
- **Multi-substrate deployment ownership:** Owned the end-to-end deployment lifecycle for onboarding the platform onto a new GovCloud substrate with stricter compliance and access restrictions — required deep cross-stack changes across Helm, Kubernetes, AWS (RDS/SQS/SNS/S3), Spinnaker, and Terraform.
- **Streaming data pipeline:** Designed the event pipeline that streams platform data into Data Cloud at 600K events/min, unlocking analytics and reporting use cases that were previously infeasible for downstream consumers.
- **Leadership & recognition:** Scrum Lead for the team; mentor to junior engineers. Winner, SDS Hackathon — configuration-driven asset ingestion with criticality scoring.

Software Development Engineer — Flipkart

July 2019 – August 2021 | Bengaluru

- **Big Billion Day readiness lead:** Owned the team's BBD scale-up; identified the hot bottlenecks, executed targeted fixes, and took the API to **2.5M requests/sec with zero major incidents** during peak — the highest-traffic event on the platform.
- **Kubernetes migration — team pilot:** Piloted one of the first services in the team to onboard onto the company-wide K8s migration; authored the deployment playbook the team adopted, shepherded subsequent services through it, and introduced HPA for demand-driven autoscaling — materially reducing deployment time and infra spend.
- **SuperCoin Zone revamp:** Led end-to-end redesign of the SuperCoin experience with a new *Progressive Milestone* construct and rewards page — shipped to drive retention and spend frequency at scale.
- **High-throughput notifications:** Designed and built the SuperCoin credit notification pipeline delivering millions of notifications daily with consistent latency and delivery guarantees.

Software Development Engineer — TiLA Labs

December 2017 – July 2019 | Bengaluru

- **Search platform:** Founding engineer on the search and discovery stack for a zero-to-one multi-language (English & Arabic) e-commerce platform; built RESTful search APIs with filter-based querying, faceted navigation, and price-range faceting.

- **Auto-suggest (ground-up rewrite):** Redesigned the auto-complete service to support multi-language inputs; measurably improved discoverability across language cohorts.
- **Category hierarchy service:** Built and owned the multi-lingual category hierarchy and metadata store.

Software Developer — ZenParent

July 2016 – October 2017 | Bengaluru

- **Contributor platform:** Designed the external-writer framework with a view-based incentive model — shipped end-to-end, drove content volume and engagement.
- **Pregnancy lifecycle framework:** Introduced a lifecycle-stage model for expecting and new parents powering a custom algorithm for stage-specific newsletters and ad targeting — improved relevance and revenue.
- **Website revamp:** Sole backend owner for the full platform revamp.
- **Search & mobile performance:** Integrated Apache Solr with WordPress for content search; rolled out Accelerated Mobile Pages (AMP) across the platform to enhance mobile user experience and SEO.

Education

Indian Institute of Technology, Madras (IIT Madras)

Chennai, 2016

Dual Degree: B. Tech in Aerospace Engineering & M. Tech in Applied Mechanics

Certifications: Salesforce Certified Scrum Lead

Skills

- **Languages & Frameworks:** Java, Python, C, JavaScript, PHP, Spring Boot, Spring Batch
- **Architecture:** Distributed Systems, Microservices, Event-Driven Design, Domain-Driven Design
- **Cloud & Infra:** AWS (RDS, SQS, SNS, S3, etc.), Terraform, Kubernetes, Docker, Helm, Spinnaker
- **Datastore:** MySQL, Redis, Kafka, Apache Solr, JPA
- **Observability:** Prometheus, Micrometer, APM, OpenTelemetry, Splunk
- **AI & Developer Tooling:** Claude Code, Cursor, MCP Servers & Skills
- **Foundations:** Data Structures & Algorithms, Object-Oriented Design, Design Patterns